



Assignment 1 - AE4238

Aero Engine Technology

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Date: August 10, 2026

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1 Problem Description

The aero engine presented is a Geared Turbofan, introducing a gearbox between the low pressure turbine and fan, enabling independent optimization for both.

The main objective of the assignment at hand is to perform a comprehensive thermodynamic and performance analysis at cruise conditions. That is, obtaining the engine performance parameters, evaluating both the component and system efficiencies, as well as analyzing the energy distribution and power requirements - All limited to **8 pages**, excluding the cover, index, and appendix.

The components of the GTF engine are, in order, a Fan, LPC, HPC, Combustion Chamber, HPT, LPT and the exit nozzles of both bypass and core.

2 Assumptions

There are several assumptions that have been taken, out of which the most basic are listed below.

- Perfect gas behavior.
- Steady State Operation
- Adiabatic Components
- Isentropic efficiencies (constant through operation, no variations).

On top of these, we also consider stoichiometric combustion products and a fuel-air ratio small enough that air properties are preserved. Moreover, the specific heats employed are considered to be constant throughout the temperature range for the sake of simplicity, even if in reality they change with temperature. We also consider the heat ratios to be constant accordingly. For the components themselves, we assume the following:

- We assume the fan to be single stage, thus having the same pressure ratio for both core and bypass flow.
- We assume no bleed air or cooling flows present in the design.
- We assume a directly coupled LPC, meaning that it lies on the low pressure shaft and not the gearbox, sharing losses only with the fan via the LPT.
- We take the nozzles to have ideal expansion, and to account for effective velocity in choked flow.

It is also worth mentioning that the fuel mass added during combustion to the core flow rate is neglected as a first approach in the definition of combustion efficiency.

For the gas power generated, it is defined to be the remaining turbine power remaining after the compression work is extracted. From how its determined, we take the assumption that the gas generator ends at the exit of the low pressure turbine, by an isentropic expansion from T_{tg} to the exit at $T_{g''}$.

3 Procedure

The calculation of pressure and temperature is done by use of the given data (listed below) going a station at a time and conserving the numbering presented in the problem statement. The use of the isentropic relation between pressure and temperature is constantly referenced, as well as the use of isentropic efficiencies for each individual device. This is, every expression where an η term is present was derived from the efficiency definition from that device. All pressure ratios are denoted by the letter π . The use of the subscript $_{id}$ refers to an isentropic conditions result, $_t$ refers to a total or stagnation property. The numbering refers to the one seen in the statement. For the results, a MATLAB code has been employed. It has been indexed in the appendix of the document for reference.

Parameter	Value
Flight Mach number, M_0	0.78
Static temperature, T_0	216.5 K
Static pressure, p_0	22 632 Pa
Total mass flow rate, $\dot{m}_{\text{total}}$	220 kg/s
Bypass ratio, BPR	12.5
Fan pressure ratio, π_{fan}	1.35
LPC pressure ratio, π_{LPC}	4.5
HPC pressure ratio, π_{HPC}	5.5
Turbine inlet temperature, T_{t4}	1600 K
Inlet pressure recovery, π_{inlet}	0.99
Combustor pressure loss, π_{comb}	0.96
Polytropic efficiencies	
Fan, LPC, HPC	0.92
HPT, LPT	0.91
Mechanical efficiency, η_{mech}	0.995
Gearbox efficiency, η_{gearbox}	0.995
Combustor efficiency, η_{comb}	0.995
Nozzle efficiency, η_{nozzle}	0.99
Gas properties	
$c_{p,\text{air}}$	1000 J/kg K
$c_{p,\text{gas}}$	1150 J/kg K
k_{air}	1.40
k_{gas}	1.33
Gas constant, R	287 J/kg K
Lower heating value, LHV	43×10^6 J/kg

Table 1: Given input data

Before analyzing any station, we compute both mass flow currents present in the turbofan:

$$\dot{m}_{\text{core}} = \frac{\dot{m}_{\text{total}}}{1 + \text{BPR}} = 16.296 \text{ kg/s}$$

$$\dot{m}_{\text{bypass}} = \text{BPR} \cdot \dot{m}_{\text{core}} = 203.704 \text{ kg/s}$$

Then, the conditions in the free stream are examined, with the use of stagnation properties and their definition:

$$v_0 = M_0 \sqrt{k_{\text{air}} R T_0} = 230.053 \text{ m/s}$$

$$T_{t0} = T_0 \left(1 + \frac{k_{\text{air}} - 1}{2} M_0^2 \right) = 242.844 \text{ K}$$

$$p_{t0} = p_0 \left(1 + \frac{k_{\text{air}} - 1}{2} M_0^2 \right)^{\frac{k_{\text{air}}}{k_{\text{air}} - 1}} = 33827.000 \text{ Pa}$$

For the compression of the flow, we first need to get the properties at 2, recognizing that the stagnation temperatures are the same for the free stream and the fan entrance:

$$p_{t2} = \pi_{\text{inlet}} p_{t0} = 33489.000 \text{ Pa}, \quad T_{t2} = T_{t0}$$

Then, for both streams at the fan:

$$p_{t13} = \pi_{\text{fan}} p_{t2} = 45210.000 \text{ Pa}$$

$$T_{t13,\text{id}} = T_{t2} \pi_{\text{fan}}^{\frac{k_{\text{air}} - 1}{k_{\text{air}}}} = 260.318 \text{ K}$$

$$T_{t13} = T_{t2} + \frac{T_{t13,\text{id}} - T_{t2}}{\eta_{\text{fan}}} = 266.476 \text{ K}$$

and moving into the core, we work out the properties after the low pressure compressor:

$$p_{t25} = \pi_{\text{LPC}} p_{t13} = 203445.000 \text{ Pa}$$

$$T_{t25} = T_{t13} + \frac{T_{t13} (\pi_{\text{LPC}}^{\frac{k_{\text{air}} - 1}{k_{\text{air}}}} - 1)}{\eta_{\text{LPC}}} = 421.973 \text{ K}$$

and after the high pressure one:

$$p_{t3} = \pi_{\text{HPC}} p_{t25} = 1118945.000 \text{ Pa}$$

$$T_{t3} = T_{t25} + \frac{T_{t25} (\pi_{\text{HPC}}^{\frac{k_{\text{air}} - 1}{k_{\text{air}}}} - 1)}{\eta_{\text{HPC}}} = 709.806 \text{ K}$$

where the overall pressure ratio is defined as: $p_{t3}/p_{t2} = 33.400$. For the combustion we can obtain:

$$p_{t4} = \pi_{\text{comb}} p_{t3} = 1074187.000 \text{ Pa}$$

$$\dot{m}_{\text{fuel}} = \dot{m}_{\text{core}} \frac{c_{p,\text{gas}} T_{t4} - c_{p,\text{air}} T_{t3}}{\eta_{\text{comb}} \cdot \text{LHV}} = 0.431 \text{ kg/s}$$

$$f = \frac{\dot{m}_{\text{fuel}}}{\dot{m}_{\text{core}}} = 0.026, \quad \dot{m}_{\text{hot}} = \dot{m}_{\text{core}} + \dot{m}_{\text{fuel}} = 16.727 \text{ kg/s}$$

where we have defined $\dot{m}_{\text{hot}}$ as the mass flow at the core with the addition of the added fuel, and we have distinguished between both the gas' properties and the air's at the entrance and exit of the device.

We can also obtain the work rates for the compression stage, knowing they will be needed to relate with the mechanical efficiency:

$$W_{\text{fan}} = \dot{m}_{\text{total}} c_{p,\text{air}} (T_{t13} - T_{t2}) = 5.199 \text{ MW}$$

$$W_{\text{LPC}} = \dot{m}_{\text{core}} c_{p,\text{air}} (T_{t25} - T_{t13}) = 2.534 \text{ MW}$$

$$W_{\text{HPC}} = \dot{m}_{\text{core}} c_{p,\text{air}} (T_{t3} - T_{t25}) = 4.691 \text{ MW}$$

And relating with the work found in the previous section, we can work out the properties after both the high and low pressure turbines:

$$W_{\text{HPT}} = \frac{W_{\text{HPC}}}{\eta_{\text{mech}}} = 4.714 \text{ MW}$$

$$T_{t45} = T_{t4} - \frac{W_{\text{HPT}}}{\dot{m}_{\text{hot}} c_{p,\text{gas}}} = 1354.897 \text{ K}$$

$$W_{\text{LPT}} = \frac{1}{\eta_{\text{mech}}} \left(\frac{W_{\text{fan}}}{\eta_{\text{gearbox}}} + W_{\text{LPC}} \right) = 7.798 \text{ MW}$$

$$T_{t5} = T_{t45} - \frac{W_{\text{LPT}}}{\dot{m}_{\text{hot}} c_{p,\text{gas}}} = 949.534 \text{ K}$$

After the turbines we can check for the state of both nozzles in the same step:

$$\left(\frac{p_{t5}}{p_0} \right)_{\text{crit,gas}} = \left(1 + \frac{k_{\text{gas}} - 1}{2} \right)^{\frac{k_{\text{gas}}}{k_{\text{gas}} - 1}} = 1.853$$

$$\left(\frac{p_{t5}}{p_0} \right)_{\text{core}} = 4.528 \quad \rightarrow \quad \text{choked, above critical value}$$

Then knowing it is choked, we can simply apply the following expression relating the temperatures and the efficiency of the nozzle. The pressures and area (with the use of the equation of state) are also computed to adapt the velocity - and thus the thrust - to the choked case:

$$T_8 = T_{t5} - \eta_{\text{nozzle}} \left(T_{t5} - \frac{T_{t5}}{1 + \frac{k_{\text{gas}} - 1}{2}} \right) = 927.028 \text{ K}$$

$$v_8 = \sqrt{2 c_{p,\text{gas}} (T_{t5} - T_8)} = 553.384 \text{ m/s}$$

$$p_8 = \frac{p_{t5}}{\left(\frac{p_{t5}}{p_0} \right)_{\text{crit,gas}}} = 55379.000 \text{ Pa}$$

$$\rho_8 = \frac{p_8}{RT_8} = 0.208 \text{ kg/m}^3, \quad A_8 = \frac{\dot{m}_{\text{hot}}}{\rho_8 v_8} = 0.145 \text{ m}^2$$

$$v_{8,\text{eff}} = v_8 + \frac{A_8(p_8 - p_0)}{\dot{m}_{\text{hot}}} = 803.711 \text{ m/s}$$

Now for the bypass flow rate, we can obtain the following pressure ratio:

$$\left(\frac{p_{t13}}{p_0}\right)_{\text{core}} = 1.998 \rightarrow \text{choked as well}$$

And we can follow the same procedure as above:

$$T_{18} = T_{t13} - \eta_{\text{nozzle}} \left(T_{t13} - \frac{T_{t13}}{1 + \frac{k_{\text{air}} - 1}{2}} \right) = 264.284 \text{ K}$$

$$v_{18} = \sqrt{2c_{p,\text{air}}(T_{t13} - T_{18})} = 296.479 \text{ m/s}$$

$$p_{18} = \frac{p_{t13}}{\left(\frac{p_{t13}}{p_0}\right)_{\text{crit,air}}} = 23884.000 \text{ Pa}$$

$$\rho_{18} = \frac{p_{18}}{RT_{18}} = 0.315 \text{ kg/m}^3, \quad A_{18} = \frac{\dot{m}_{\text{bypass}}}{\rho_{18}v_{18}} = 2.182 \text{ m}^2$$

$$v_{18,\text{eff}} = v_{18} + \frac{A_{18}(p_{18} - p_0)}{\dot{m}_{\text{bypass}}} = 307.833 \text{ m/s}$$

With the effective velocities at both nozzles we can now obtain the corresponding thrust and specific fuel consumption as follows:

$$F_{\text{total}} = \dot{m}_{\text{hot}}(v_{8,\text{eff}} - v_0) + \dot{m}_{\text{bypass}}(v_{18,\text{eff}} - v_0) = 25.440 \text{ kN}$$

$$\text{SFC} = \frac{\dot{m}_{\text{fuel}}}{F_{\text{total}}} \times 3600 = 0.061 \text{ kg/(N}\cdot\text{h)} = 16.920 \text{ g/(kN}\cdot\text{s)}$$

4 Results and Observations

4.1 Efficiency calculations

For the efficiency calculations, we can simply follow the provided formulas, taking into account that the velocity at the exit of the engine is the effective one, as well as taking both core and bypass flows into account with the summations:

$$\eta_{\text{comb}} = \frac{\dot{m}_{\text{hot}}(c_{p,\text{gas}}T_{t4} - c_{p,\text{air}}T_{t3})}{\dot{m}_{\text{fuel}} \text{LHV}_f} = 99.500\%$$

$$\eta_{\text{thdy}} = \frac{W_{\text{gg}}}{\dot{m}_{\text{hot}}(c_{p,\text{gas}}T_{t4} - c_{p,\text{air}}T_{t3})} = 57.740\%$$

$$\eta_{\text{jet, gener.}} = \frac{\sum \left\{ \frac{1}{2} \dot{m} (v_{j,eff}^2 - v_0^2) \right\}}{W_{gg}} = 84.480\%$$

$$\eta_{\text{prop}} = \frac{\sum \{ \dot{m} (v_{j,eff} - v_0) \} v_0}{\sum \left\{ \frac{1}{2} \dot{m} (v_{j,eff}^2 - v_0^2) \right\}} = 63.470\%$$

$$\eta_{\text{thermal}} = \frac{\sum \left\{ \frac{1}{2} \dot{m} (v_{j,eff}^2 - v_0^2) \right\}}{\dot{m}_{fuel} \text{LHV}_f} = 49.810\%$$

$$\eta_{\text{total}} = \frac{\sum \{ \dot{m} (v_{j,eff} - v_0) \} v_0}{\dot{m}_{fuel} \text{LHV}_f} = 31.620\%$$

Where the term W_{gg} corresponds to the gas power produced by the geared turbofan, defined with:

$$T_{tg} = T_{t45} - \frac{1}{\dot{m}_{hot} \cdot c_{p,gas}} \left(\frac{W_{fan} \cdot \dot{m}_{core} / \dot{m}_{bypass}}{\eta_{gearbox} \cdot \eta_{mech}} + \frac{W_{LPC}}{\eta_{mech}} \right) = 1200.689 \text{ K}$$

$$P_{tg} = P_{t45} \left[1 - \frac{1 - T_{tg}/T_{t45}}{\eta_{LPT}} \right]^{\frac{k_{gas}}{k_{gas}-1}} = 298242.000 \text{ Pa}$$

$$T_{8''} = T_{tg} \left(\frac{P_{tg}}{P_0} \right)^{\frac{1}{k_{gas}} - 1} = 633.248 \text{ K}$$

$$W_{gg} = \dot{m}_{hot} c_{p,gas} (T_{tg} - T_{8''}) = 10.915 \text{ MW}$$

4.2 Sankey diagram

Once the efficiencies are determined, it proves straightforward to compute the corresponding Sankey diagram:

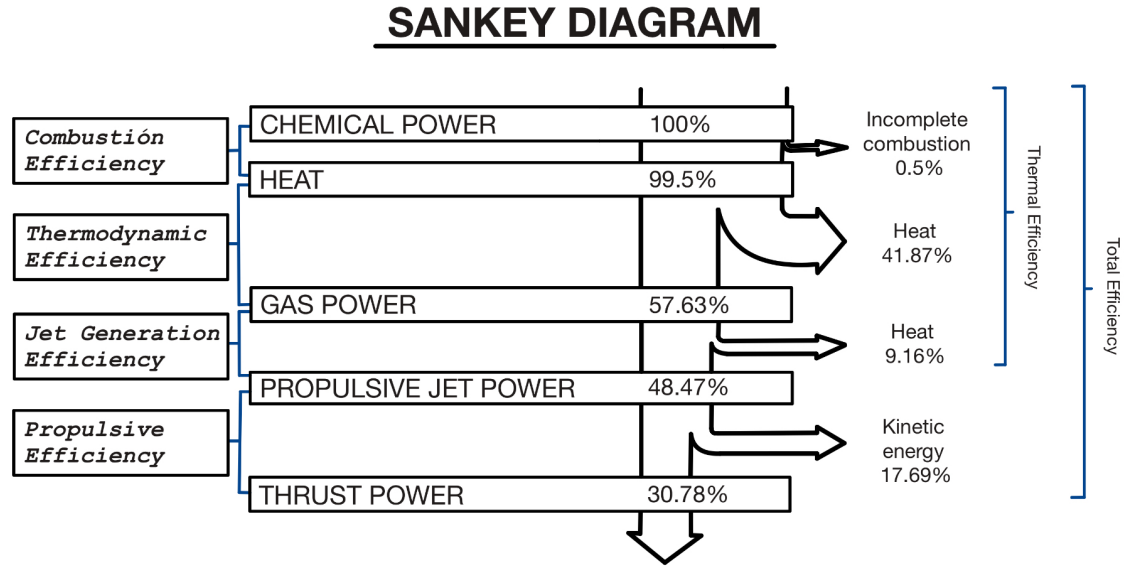


Figure 1: Sankey Diagram for the Geared Turbofan

The generated Sankey diagram provides a clear breakdown of the energy cascade within the Geared Turbofan. A key observation is the contrast between the minimal combustion losses of 0.5% and the substantial heat rejection to the atmosphere of 41.87% due to the low Thermodynamic Efficiency. However, the drop to a final Propulsive Jet Power of 48.47% and an Overall Efficiency of 30.78% shows the magnitude of the kinetic energy wasted in the exhaust wake of 17.69%, identifying this as a primary area for potential optimization through bypass ratio enhancements, we will talk about this later on in the possible improvements.

4.3 Power available for the compressor

To obtain the available power for compression right after the combustion chamber, one may realize that, by definition, it will be the difference between the power generated in the combustion chamber with that of the gas power generated, as it simply is the amount that is left to be taken back to the compression stage through the shaft:

$$W_{available,comp} = \dot{m}_{hot} (c_{p,gas} \cdot T_{t4} - c_{p,air} \cdot T_{t3}) - W_{gg} = 7.989 \text{ MW}$$

5 Conclusions and Outlook

The following conclusions have been drawn from the exercise:

- **Thrust Generated:** The engine produces a total thrust of 25.42 kN at cruise conditions, being comprised in a 62% of bypass mass flow, as expected from a modern turbofan.
- **Thermal Efficiency:** Lying just below 50%, one can deduce that half of the fuel's chemical energy is lost before being converted into kinetic energy, mainly due to irreversibilities and exhaust heat.

- **Propulsive efficiency:** Its value of 63.47% indicates large energy losses in exhaust jets, representing the main limiting factor in the engine performance.
- **Choked Nozzles:** Both the core and bypass nozzles operate under choked conditions, as is expected for cruise conditions and the high pressure ratio induced from the ambient.
- **Gearbox implementation:** It allows the fan to exist decoupled from the low pressure turbine, introducing a 0.5% power loss. This power loss is negligible, and the performance gains justify it.
- **Underexpanded Flow:** The core nozzle has a pressure ratio of 4.528, indicating incomplete expansion which wastes work potential. This has likely been done to conserve a very high overall pressure ratio of 33.4, though it means reserving 7.989 MW for compression power.
- **SFC value:** The value achieved is quite low, reflecting the good fuel efficiency for this kind of engine.

5.1 Possible improvements on losses and efficiency

To reduce losses and improve overall efficiency, several strategies could be implemented, of which a couple will be mentioned. First, further increasing the bypass ratio would reduce the average exhaust velocity, improving the propulsive efficiency by better matching jet velocity to flight speed and reducing the wasted kinetic energy losses. This creates challenges in engine weight and induced shape drag, thus notably increasing the costs. Second, since both nozzles are currently choked and underexpanded (as evidenced by the core, with $\text{NPR} = 4.53$), replacing the purely convergent nozzles with a convergent-divergent alternative would allow the exhaust gases to expand fully to ambient pressure, recovering the pressure thrust more efficiently and reducing losses due to expansion. If implemented, the core would benefit much more from this than the bypass flow. Finally, increasing the overall pressure ratio above 33.4 and raising the turbine inlet temperature above 1600 K would improve thermodynamic efficiency, though this would necessitate advanced materials (such as ceramic matrix composites, CMCs) and improved cooling schemes to maintain the turbine's lifespan. Out of all, the most impactful improvements would be increasing BPR and implementing CD nozzles, as they directly address the two largest loss mechanisms visible in the Sankey diagram without too costly an implementation or too big an advancement.

There are other alternatives like the use of an inter cooler or improvement of individual component efficiencies or shaft-like components, not mentioned because of proving more detrimental than advantageous, and because of the assumption that their value cannot be reasonably increased, respectively.

6 Appendix - Code

```
1 %% AE4238 Assignment 1 - GTF Analysis
2
3 clear all; clc;
4
5 %% Given Parameters
6 M0 = 0.78; T0 = 216.5; P0 = 22632;
7 pi_inlet = 0.99; m_dot_total = 220; BPR = 12.5;
8 pi_fan = 1.35; pi_LPC = 4.5; pi_HPC = 5.5;
9 T_t4 = 1600;
10
11 eta_fan = 0.92; eta_LPC = 0.92; eta_HPC = 0.92;
12 eta_LPT = 0.91; eta_HPT = 0.91;
13 eta_mech = 0.995; eta_gearbox = 0.995;
14 eta_comb = 0.995; pi_comb = 0.96; eta_nozzle = 0.99;
15
16 R = 287;
17 cp_air = 1000; k_air = 1.4;
18 cp_gas = 1150; k_gas = 1.33;
19 LHV = 43e6;
20
21 gamma_air = (k_air - 1) / k_air; %% Just to make it quicker
    when computing isentropic relations
22 gamma_gas = (k_gas - 1) / k_gas;
23
24 %% Mass Flow Split
25 m_dot_core = m_dot_total / (1 + BPR);
26 m_dot_bypass = BPR * m_dot_core;
27
28 fprintf('=== MASS FLOW RATES ===\n');
29 fprintf('Total: %.2f kg/s | Core: %.2f kg/s | Bypass: %.2f kg
    /s\n\n', ...
30     m_dot_total, m_dot_core, m_dot_bypass);
31
32 %% Station 0 - Freestream
33 V0 = M0 * sqrt(k_air * R * T0);
34 T_t0 = T0 * (1 + (k_air-1)/2 * M0^2);
35 P_t0 = P0 * (1 + (k_air-1)/2 * M0^2)^(k_air/(k_air-1));
36
37 %% Compression Stages
38 P_t2 = pi_inlet * P_t0; T_t2 = T_t0;
39
40 % Fan (both streams)
41 P_t13 = pi_fan * P_t2;
42 T_t13_id = T_t2 * pi_fan^gamma_air;
43 T_t13 = T_t2 + (T_t13_id - T_t2) / eta_fan;
44 T_t21 = T_t13; P_t21 = P_t13;
45
```

```

46 % LPC (core only)
47 P_t25 = pi_LPC * P_t21;
48 T_t25_id = T_t21 * pi_LPC^gamma_air;
49 T_t25 = T_t21 + (T_t25_id - T_t21) / eta_LPC;
50
51 % HPC
52 P_t3 = pi_HPC * P_t25;
53 T_t3_id = T_t25 * pi_HPC^gamma_air;
54 T_t3 = T_t25 + (T_t3_id - T_t25) / eta_HPC;
55
56 fprintf('=== COMPRESSION ===\n');
57 fprintf('T_t2=%.1fK, T_t13=%.1fK, T_t25=%.1fK, T_t3=%.1fK\n',
58         T_t2, T_t13, T_t25, T_t3);
59 fprintf('Overall PR = %.1f\n\n', P_t3/P_t2);
60
61 %% Combustion
62 P_t4 = pi_comb * P_t3;
63 m_dot_fuel = m_dot_core * (cp_gas*T_t4 - cp_air*T_t3) / (
64     eta_comb*LHV);
65 f = m_dot_fuel / m_dot_core;
66 m_dot_hot = m_dot_core + m_dot_fuel;
67
68 fprintf('=== COMBUSTION ===\n');
69 fprintf('Fuel flow: %.4f kg/s | f = %.5f\n\n', m_dot_fuel, f)
70 ;
71
72 %% Compressor Work
73 W_fan = m_dot_total * cp_air * (T_t13 - T_t2);
74 W_LPC = m_dot_core * cp_air * (T_t25 - T_t21);
75 W_HPC = m_dot_core * cp_air * (T_t3 - T_t25);
76
77 %% Turbines
78 % HPT drives HPC
79 W_HPT = W_HPC / eta_mech;
80 dT_HPT = W_HPT / (m_dot_hot * cp_gas);
81 T_t45 = T_t4 - dT_HPT;
82 T_t45_id = T_t4 - (T_t4 - T_t45) / eta_HPT;
83 pi_HPT = (T_t45_id / T_t4)^(k_gas/(k_gas-1));
84 P_t45 = pi_HPT * P_t4;
85
86 % LPT drives Fan + LPC
87 % W_LPT = (W_fan + W_LPC) / (eta_mech * eta_gearbox);
88 W_LPT = (W_fan / eta_gearbox + W_LPC) / eta_mech;
89 dT_LPT = W_LPT / (m_dot_hot * cp_gas);
90 T_t5 = T_t45 - dT_LPT;
91 T_t5_id = T_t45 - (T_t45 - T_t5) / eta_LPT;
92 pi_LPT = (T_t5_id / T_t45)^(k_gas/(k_gas-1));
93 P_t5 = pi_LPT * P_t45;

```

```

192 fprintf('== TURBINES ==\n');
193 fprintf('HPT: W=%.2fMW, T_t45=%.1fK\n', W_HPT/1e6, T_t45);
194 fprintf('LPT: W=%.2fMW, T_t5=%.1fK, P_t5=%.0fPa\n\n', W_LPT/1
    e6, T_t5, P_t5);
195
196 %% Core Nozzle
197 NPR_crit_gas = (1 + (k_gas-1)/2)^(k_gas/(k_gas-1));
198 NPR_core = P_t5 / P0;
199
200 if NPR_core > NPR_crit_gas
201     % CHOKED
202     M8 = 1.0;
203     T8_id = T_t5 / (1 + (k_gas-1)/2);
204     T8 = T_t5 - eta_nozzle*(T_t5 - T8_id);
205     V8 = sqrt(2*cp_gas*(T_t5 - T8));
206     P8 = P_t5 / NPR_crit_gas;
207     rho8 = P8 / (R*T8);
208     A8 = m_dot_hot / (rho8*V8);
209     V8_eff = V8 + A8*(P8 - P0)/m_dot_hot; % Defined by
    solving for effective velocity in the thrust equation
210     core_choked = true;
211 else
212     % UNCHOKED
213     P8 = P0;
214     T8_id = T_t5 * (P8/P_t5)^gamma_gas;
215     T8 = T_t5 - eta_nozzle*(T_t5 - T8_id);
216     V8 = sqrt(2*cp_gas*(T_t5 - T8));
217     V8_eff = V8; % Idem
218     A8 = 0;
219     core_choked = false;
220 end
221
222 %% Bypass Nozzle
223 NPR_crit_air = (1 + (k_air-1)/2)^(k_air/(k_air-1));
224 NPR_bypass = P_t13 / P0;
225
226 if NPR_bypass > NPR_crit_air
227     % CHOKED
228     M18 = 1.0;
229     T18_id = T_t13 / (1 + (k_air-1)/2);
230     T18 = T_t13 - eta_nozzle*(T_t13 - T18_id);
231     V18 = sqrt(2*cp_air*(T_t13 - T18));
232     P18 = P_t13 / NPR_crit_air;
233     rho18 = P18 / (R*T18);
234     A18 = m_dot_bypass / (rho18*V18);
235     V18_eff = V18 + A18*(P18 - P0)/m_dot_bypass;
236     bypass_choked = true;
237 else
238     % UNCHOKED

```

```

139     P18 = P0;
140     T18_id = T_t13 * (P18/P_t13)^gamma_air;
141     T18 = T_t13 - eta_nozzle*(T_t13 - T18_id);
142     V18 = sqrt(2*cp_air*(T_t13 - T18));
143     V18_eff = V18;
144     A18 = 0;
145     bypass_choked = false;
146 end
147
148 fprintf('=== NOZZLES ===\n');
149 fprintf('Core: V8=%.1fm/s, P8=%.0fPa, Choked=%d\n', V8, P8,
        core_choked);
150 fprintf('   Core effective velocity: V8_eff = %.1f m/s\n',
        V8_eff);
151 fprintf('Bypass: V18=%.1fm/s, P18=%.0fPa, Choked=%d\n', V18,
        P18, bypass_choked);
152 fprintf('   Bypass effective velocity: V18_eff = %.1f m/s\n\n',
        , V18_eff);
153
154 %% Thrust Calculation
155
156 % **Using effective jet velocities**
157 F_core_eff = m_dot_hot * (V8_eff - V0);
158 F_bypass_eff = m_dot_bypass * (V18_eff - V0);
159 F_total = F_core_eff + F_bypass_eff;
160
161 fprintf('=== THRUST ===\n');
162 fprintf('Core: %.2f kN | Bypass: %.2f kN | TOTAL: %.2f kN\n',
        ...
163     F_core_eff/1e3, F_bypass_eff/1e3, F_total/1e3);
164
165
166 %% SFC
167 SFC = m_dot_fuel / F_total;
168 SFC_hr = SFC * 3600; % Units to kg/N.hour since it is also
        very commonly used
169
170 fprintf('=== SFC ===\n');
171 fprintf('SFC = %.5f kg/(N.hr) = %.2f g/(kN.s)\n\n', SFC_hr,
        SFC*1e6);
172
173 %% EFFICIENCIES
174
175 fprintf('=== COMPLETE EFFICIENCY ANALYSIS ===\n\n');
176
177 % Chemical energy input from fuel
178 Q_chem = m_dot_fuel * LHV;
179

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180 % Heat released in combustor (accounting for combustor
    efficiency)
181 Q_released = eta_comb * m_dot_fuel * LHV;
182
183 % Heat added to working fluid
184 Q_heat_from_comb = m_dot_hot * (cp_gas * T_t4 - cp_air * T_t3
    ); % Heat from combustion
185
186 % Gas generator power (with gas parameters)
187 T_tg = T_t45 - ((W_fan*m_dot_core/m_dot_bypass)/(eta_gearbox*
    eta_mech) + W_LPC/eta_mech)/(m_dot_hot*cp_gas);
188 P_tg = P_t45*(1 - ((1-T_tg/T_t45)/(eta_LPT)))^(k_gas/(k_gas
    -1));
189 T_8dprime = T_tg*(P_tg/P0)^(1/k_gas - 1); % Temp. double
    prime
190 W_gg = m_dot_hot*cp_gas*(T_tg - T_8dprime); % Gas power
191
192 % Kinetic energy produced in jets
193 KE_core = 0.5*m_dot_hot*(V8_eff^2 - V0^2);
194 KE_bypass = 0.5*m_dot_bypass*(V18_eff^2 - V0^2);
195 Delta_KE = KE_bypass + KE_core;
196
197 % Thrust power
198 W_thrust = F_total * V0;
199
200
201 % === EFFICIENCY DEFINITIONS ===
202
203 fprintf('--- Efficiencies ---\n');
204
205 % COMBUSTOR EFFICIENCY (given)
206 eta_comb_used = eta_comb;
207 fprintf('Combustor efficiency:          eta_comb = %.4f (%.2f
    %%) [Given]\n', ...
208     eta_comb_used, eta_comb_used*100);
209
210 % THERMODYNAMIC EFFICIENCY
211 eta_thdy = W_gg / Q_heat_from_comb;
212 fprintf('Thermodynamic efficiency:      eta_thdy = %.4f (%.2f
    %%) \n', eta_thdy, eta_thdy*100);
213
214 % JET GENERATION EFFICIENCY
215 eta_jet_gener = Delta_KE / W_gg;
216 fprintf('Jet generation efficiency:      eta_jet = %.4f (%.2f
    %%) \n', eta_jet_gener, eta_jet_gener*100);
217
218 % PROPULSIVE EFFICIENCY
219 eta_prop = W_thrust / Delta_KE;

```

```

220 fprintf('Propulsive efficiency:          eta_prop = %.4f (%.2f
      %%)\n', eta_prop, eta_prop*100);
221
222 % THERMAL EFFICIENCY
223 eta_th = Delta_KE / Q_chem;
224 fprintf('Thermal efficiency:          eta_th   = %.4f (%.2f
      %%)\n', eta_th, eta_th*100);
225
226 % OVERALL/TOTAL EFFICIENCY
227 eta_total = W_thrust / Q_chem;
228 fprintf('Overall/Total efficiency:    eta_ov   = %.4f (%.2f
      %%)\n', eta_total, eta_total*100);
229
230
231
232 W_available_comp = m_dot_hot*(cp_gas*T_t4- cp_air*T_t3) -
      W_gg;
233 fprintf('\n--- Remaining Power for Compression---\n');
234 fprintf('  Remaining power for combustion: W_available_comp =
      %.3f MW\n', W_available_comp/1e6);

```